Adapting the Interface, Not the Model

Life-Harness: runtime harness adaptation for deterministic LLM agents · Tianshi Xu, Huifeng Wen, Meng Li · Peking University · arXiv 2605.22166v3

HarnessRSI 72 ★★★

evidence: retained

Paper

Code

1 What changes, what is kept

At test time: the harness wraps a frozen model

1 **Environment Contract** — clearer tool descriptions, policy limits, common pitfalls

2 **Procedural Skill** — BM25 top-1 skill from training trajectories → system prompt

Frozen LLM (Qwen3-4B-Instruct) · weights never change

3 **Action Realization** — check, canonicalize or block the action before execution

Environment (tasks, tools, scorer unchanged)

4 **Trajectory Regulation: loops, stalls → recovery message**

Offline evolution: 6 rounds, training tasks only

a **Run frozen solver** — current harness on ≤ 50 training tasks per round → trajectories and scores

b **Evolution LLM (GPT-5.5)** — finds recurring failures; adds, revises or removes mechanisms in the 4 layers

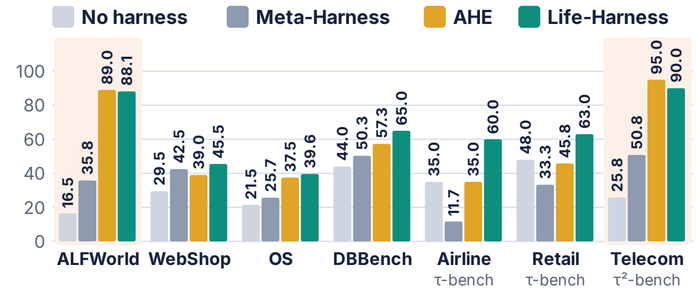
c **Validate** — interface checks + trace replay; failures go back, ≤ 3 repair attempts

d **Keep or reject** — same training batch, benchmark scorer: replace only if accuracy strictly improves

6×

retained harness frozen → held-out test tasks
one harness per environment; reused on other models

2 Headline: big gain over no harness



Qwen3-4B-Instruct, test split, Pass@1 %, one evolved harness per environment. Airline, Retail: τ-bench; Telecom: τ²-bench (20 / 40 / 40 test tasks). Shaded: AHE higher.

3 Controls that decide if the gain is real

Layer ablation (Table 4, single runs)

rel. drop, %	Air	Ret	Tel	ALF	Web	OS	DB
w/o Contract	-8.3	-17.5	-16.0	-1.0	-4.4	-14.1	-16.9
w/o Skill	-8.3	-15.9	-17.4	-1.0	-2.2	-14.1	-3.1
w/o Action	-61.7	-15.9	-10.1	-1.0	-6.6	-59.6	-4.6
w/o Trajectory	-3.3	-16.7	-36.2	-86.5	-26.4	-14.1	-4.6

Relative drop vs full harness, Qwen3-4B. No SD; re-evolved or layer simply removed is not stated.

Vs other harness evolvers (Table 1)

Qwen3-4B	AgentBench	τ Pass@1	tokens 10^7
No harness	27.9	36.3	–
Meta-Harness	38.6	**31.9**	3.3 · 3.0
AHE	55.7	58.6	**9.9 · 8.2**
Life-Harness	**59.5**	**71.0**	**2.5 · 3.5**

Authors' runs of open-source code (AHE built for coding agents); same rounds and batches, tokens not matched. Tokens: AgentBench · τ.

Same harness, other models

- 17 more open-weight models: 116 of 126 improve
- GPT 5.5 / Opus 4.8 / Gemini 3.1 Pro: mean +2.8 / +4.4 / +3.4 pp, 18 of 21
- AppWorld (57 tasks): mean 25.5 → 30.8, 16 of 18 models
- ALIGN / AgentSquare: released harnesses, ALIGN on its own task split

4 Where the gain is absent or thin

- **AHE is higher** on ALFWorld (89.0 vs 88.1) and Telecom (95.0 vs 90.0)
- Meta-Harness is **below the unmodified model** on Airline (11.7 vs 35.0), Retail (33.3 vs 48.0)
- More tokens than Meta-Harness on WebShop, Retail, Telecom; 8.5% saving is aggregate
- Gains **uneven**: Qwen3.5-27B Airline −3.3, ALFWorld 0.0, WebShop +7.5; 10 of 126 not improved
- Prompt-only (Fig. 6, approx.): at or below unmodified on ALFWorld, OS
- **No SD or CI**, no repeats reported; frozen round not stated

Takeaway

Evolving four interface layers from training failures lifts a frozen 4B agent a lot (τ Pass@1 mean 36.3 → 71.0) and carries over to other models. Against AHE the margin is 7.5 pp on single runs, AHE wins two environments, and gains are uneven across models and environments.

Our read · HarnessRSI

Score 72 · ★★★ · evidence level: retained

Reports: held-out test, 18 backbones, layer ablations, cost.
Does not report: run variance, cross-environment transfer, leakage audit.